# MILITARY STANDARD

**MIL-STD-1386-21 A**

**(NAVY)**

14 MARCH 1977

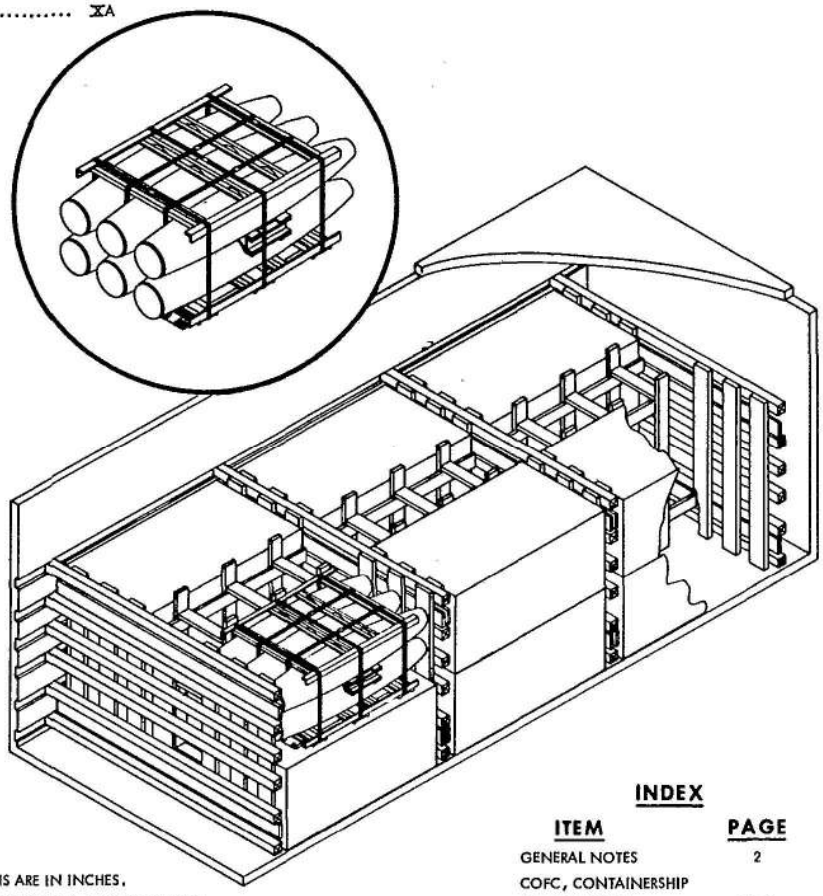
SUPERSEDING
MIL-STD-1386-21 (NAVY)
1 JUNE 1973

## LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

### BOMB,500LB MK 82 MOD 2 (THERMALLY PROTECTED) ON PALLET, MHU-122/E FLEET ISSUE UNIT LOAD

**UNIT LOAD DATA**

| | |
|---|---|
| UNIT LOAD DWG | WR-54/239 |
| WEIGHT OF UNIT LOAD (APPROX) | 3165 LBS |
| WEIGHT OF PALLET (BOTTOM FRAME) | 113 LBS |
| DIMENSIONS | 61 3/8 L X 35W X 29 H |
| CUBE | 35.9 CU FT |
| DOT HAZARD CLASSIFICATION | EXPLOSIVES A |
| COAST GUARD CLASS | XA |



NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS, REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

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**AUTHORIZED FOR CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENT.**

M.S. Gray NAVHC 2/25/77
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J.E. Kelly 2/28/77
SIGNATURE AIR SYSCOM, BY DIRECTION DATE

**APPROVED BY US COAST GUARD**

P.A. Atterno
DATE 22 Feb 1977

**APPROVED BY BUREAU OF EXPLOSIVES**

A.F. Brandstadt
DATE 10/20/75

**ORIGINATOR** S.R. Petoia 8025 2 Feb 77
SIGNATURE

NAVAL WEAPONS HANDLING CENTER
WPNSTA EARLE, NEW JERSEY



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# GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF MK 82 MOD 2 500 LB THERMAL COATED BOMBS. THE BOMBS SHALL BE PALLETIZED IN ACCORDANCE WITH WR-54/239.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6 C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6 C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.

**CAUTION**

WHEN POSITIONING CROSSMEMBERS, MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO BELT RAILS FOR SHIPMENT.
5. WHEN THE VOID BETWEEN THE CROSSMEMBER AND THE SEPARATOR GATE IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBERS AND GATES AS SHOWN IN DETAIL D.
6. THE FILLER ASSEMBLY, DETAIL C, IS PREASSEMBLED AND SLID INTO PLACE WITH BOTTOM STOP, PIECE 7, ON THE RIGHT SIDE OF THE MILVAN. THE STRUT PIECE 5, SHALL BE OF SUFFICIENT LENGTH TO CAUSE THE ASSEMBLY TO FILL THE VOID BETWEEN THE UNIT LOADS. ONE INCH SLACK IS PERMITTED.

**WARNING**

THE FILLER ASSEMBLY MUST BE ORIENTED AS DESCRIBED ABOVE.

7. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC. MM-L-751
   NAILS - FED. SPEC. FF-N-105, TYPE II, CLASS 10, COMMON BRIGHT.
8. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA, CLOSE AND SEAL CONTAINER DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. 1).

# COFC, CONTAINERSHIP AND HIGHWAY

1. THIS CONTAINER LOAD CONSISTS OF 12 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS DOCUMENT.
2. THE LOADED MILVAN CONTAINER MAY BE MOVED OVER PUBLIC HIGHWAYS USING COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY THAT MEETS THE REQUIREMENTS OF THE CODE OF FEDERAL REGULATIONS.

**WARNING**

BECAUSE THE WEIGHT LOADED INTO THIS CONTAINER EXCEEDS 34,000 LBS, THE MIVAN CHASSIS CANNOT BE USED TO TRANSPORT THE LOADED CONTAINER ON PUBLIC HIGHWAYS. THE LOADED CONTAINER MAY BE TRANSPORTED OVER DEPOT ROADS USING A MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES OR COMMERCIAL EQUIPMENT OF APPROPRIATE CAPACITY.

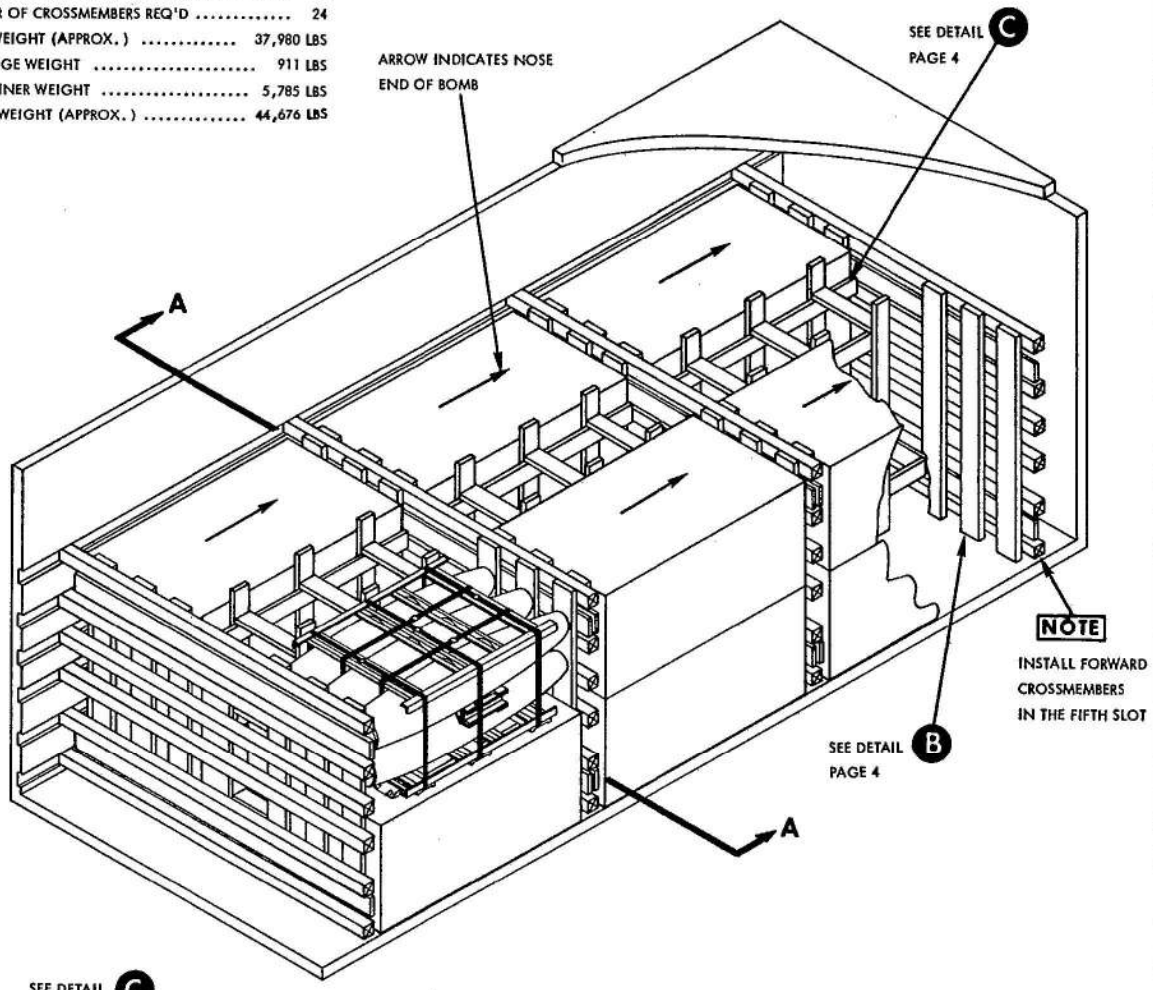
3. WHEN SHIPPING THE LOADED MILVAN CONTAINER COFC, THE FLATCAR SHALL MEET THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6C.
4. POSITION SEPARATOR GATES, DETAIL B, WITH VERTICAL MEMBERS, PIECE 1, AGAINST THE UNIT LOADS.

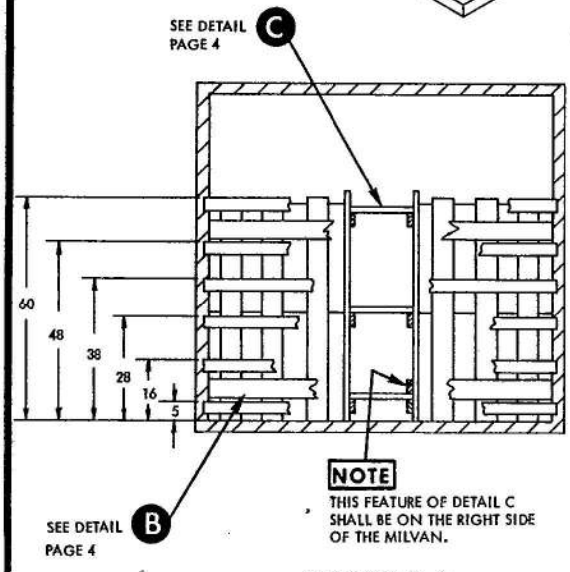


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## CONTAINER LOAD DATA

| | |
|---|---|
| NUMBER OF UNIT LOADS | 12 |
| NUMBER OF CROSSMEMBERS REQ'D | 24 |
| LOAD WEIGHT (APPROX.) | 37,980 LBS |
| DUNNAGE WEIGHT | 911 LBS |
| CONTAINER WEIGHT | 5,785 LBS |
| GROSS WEIGHT (APPROX.) | 44,676 LBS |



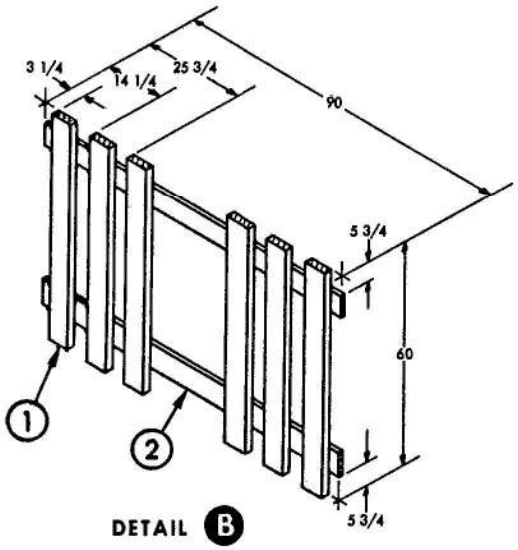

SECTION A-A

| PIECE NO. | DESCRIPTION | SIZE | NO. PCS. REQ'D | NAIL TO | NAILS: NUMBER | NAILS: SIZE |
|---|---|---|---|---|---|---|
| 7 | BOTTOM STOP | 2 X 4 X 64 | 3 | SEE 3 | - | - |
| 6 | TOP STOP & LONGITUDINAL | 2 X 4 X 64 | 3 | SEE 3 | - | - |
| 5 | CLEAT | 2 X 4 X TO SUIT | 27 | TOENAIL TO 3 | 2 PER JOINT | 10d |
| 4 | LONGITUDINAL | 2 X 4 X 42 1/2 | 15 | SEE 3 | - | - |
| 3 | UPRIGHT | 2 X 4 X 62 | 18 | 4 | 3 PER JOINT | 10d |
| 2 | HORIZONTAL | 1 X 6 X 90 | 12 | SEE 1 | 3 PER JOINT | 10d |
| 1 | VERTICAL | 2 X 6 X 60 | 36 | 2, 6 OR 7 | 3 PER JOINT | 10d |

LIST OF MATERIALS AND NAILING DATA

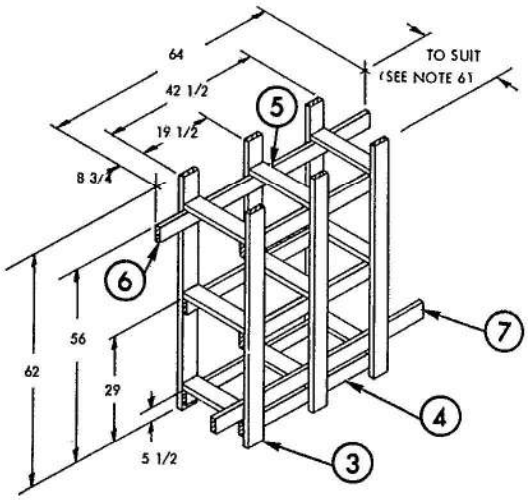


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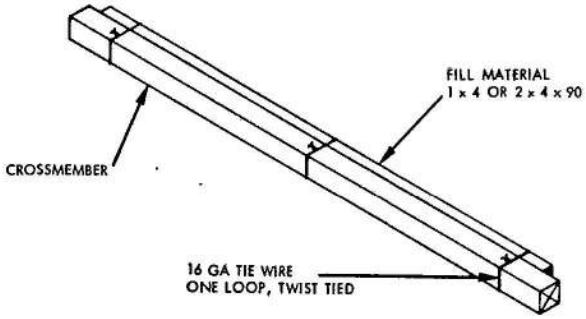

**DETAIL B**

SEPARATOR GATE
6 REQUIRED
(SEE COFC NOTE 4)
POSITION VERTICALS, PIECE 1,
AGAINST UNIT LOAD




**DETAIL C**

FILLER ASSEMBLY
3 REQUIRED
SEE GENERAL NOTE 6




**DETAIL D**

FILL MATERIAL
SEE GENERAL NOTE 5

REVIEW ACTIVITY:
NAVY - OS, AS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N315)

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